Tanish Patil

MATHEMATICS / FINANCE / COMPUTER SCIENCE

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PROFILE

Builder and educator working across quantitative finance, software, robotics and community programs. Public Python/Streamlit applications and a browser-based research suite, alongside hands-on engineering and teaching.

SELECTED ENGINEERING & RESEARCH EXPERIENCE

ThinkNeuro Internship

Conducted computational neurology research; analyzed brain data and brain-wave patterns; coded in R; created datasets, figures, a research poster, and a multipage report.

NC TSA Animatronics

Built a 4x4 ft interactive house using pneumatics, Arduino boards, sensors, and motors; placed 3rd in the state and achieved a top-10 global placement in animatronics.

FRC Robowhales

Electrical/Marketing Subteam Member & Outreach Manager; repaired the robot elevator, wired servo motors, organized outreach, and helped secure sponsors; team won THOR East and became sector champions.

Building U Coding Internship

Improved website functionality using JavaScript; progressed from Scratch to Python simulations; redesigned the website with guidance from software engineers.

Young Investors Club

Vice President; taught investing and financial planning; organized a stock pitch competition.

PUBLIC SOFTWARE

Quant Finance Python Projects

Public repository documenting portfolio optimization, risk, derivatives, fixed income, statistical arbitrage, market regimes and NLP applications built with Python and Streamlit.

Quant Research Systems

Ten modular browser research demonstrations using deterministic synthetic data. JavaScript simulations, shared mathematical primitives, bounded inputs and Canvas/table outputs.

TOOLS

Python, JavaScript, R, Arduino, Onshape, Unity. Finance, financial modeling, computational research, robotics, mathematics and teaching.

Tanish Patil

SELECTED LEADERSHIP, SERVICE & RECOGNITION

STEMPulse

Founder & President; grew membership from 7 to 20+; led STEM projects, design challenges, discussions, podcasts, and tutoring opportunities.

MathCounts

Founded a 25+ student chapter; served as 1 of 10 national community coaches; created lessons and coached students; received a coaching scholarship.

Youth Ambassadors of Service

Executive Assistant; helped fundraise $10,000+; managed social media with 18,000 views/month and 400 new followers in 3 months.

District C Innovation Internship

Helped EatWellExchange grow its volunteer base from 170 to 300+; created email templates and volunteer training; redesigned website and social media.

Math Olympiad

Three-time Math Olympiad champion; competed in open-ended mathematical problem solving.

Abacus

Graduated Abacus Level 18.

ADDITIONAL RECOGNITION

Three-time Math Olympiad champion. DECA NC State Proficiency Award; 6th of 82 in Marketing Communications. Onshape CAD certification. Abacus Level 18.

PROGRAMS & CURRENT LEADERSHIP

Youth Economy Lab - Financial Modeling Track; QOSI Research Fellowship - Machine Learning; HBA - Finance in Healthcare; C2S Tech Internship; FinU Finance Academy; TSA Treasurer; Young Investors Society VP; FLL Wizistors Lead Coach; FTC Second Coach; Code Wiz Coding Instructor.

Selected profile based on supplied achievements and public repository documentation. Program dates and academic information were not supplied. Prospective opportunities are excluded.